Linear Regression: Assumptions and Related Concepts

Detailed 5-Page Study Report

This report explains the five major assumptions behind ordinary least squares (OLS) linear regression and four closely related topics. The focus is conceptual understanding, model reliability, and practical interpretation.

1. Introduction to Linear Regression

Linear regression is a supervised learning method used to model the relationship between a dependent variable and one or more independent variables. In simple linear regression, the model estimates a straight-line relationship between an input X and an output Y.

The general form is: **Y =** β■ **+** β■**X +** ε, where β■ is the intercept, β■ is the slope, and ε represents the random error.

The central idea of OLS is to choose coefficients that minimize the sum of squared differences between observed and predicted values. The assumptions discussed in this report are important because violations can make coefficient estimates, standard errors, confidence intervals, or statistical conclusions unreliable.

2. Why Assumptions Matter

Regression assumptions are not arbitrary rules. They describe conditions under which the mathematical and statistical properties of OLS work as intended. Some assumptions concern the relationship between variables, while others concern the behavior of the errors.

• They help determine whether a linear model is appropriate.

• They affect the reliability of coefficient estimates and inference.

• They provide a framework for diagnosing model problems.

• They help identify when transformations, additional variables, or alternative models may be needed.

The Five Main Assumptions

• Linearity

• Independence of observations/errors

• Homoscedasticity

• Normality of residuals

• No perfect multicollinearity

3. Assumption 1 — Linearity

The linearity assumption means that the expected value of the dependent variable has a linear relationship with the predictors. In simple regression, the relationship is represented by a straight line.

This does not necessarily mean that every individual observation must lie on a straight line. Real data naturally contain noise. The important idea is that the systematic part of the relationship should be adequately represented by a linear function.

Why it matters

If the true relationship is strongly curved but a straight-line model is used, the model can systematically underpredict some regions and overpredict others. This can produce structured residual patterns and misleading coefficients.

How to assess it

• Inspect a scatter plot of the outcome against important predictors.

• Examine residuals versus fitted values.

• Look for systematic curves or patterns in residual plots.

• Consider transformations or nonlinear terms when justified by the data and domain.

Assumption 2 — Independence

Independence means that the observations, or more precisely the relevant errors, are not correlated with one another. One observation should not provide information about another observation's unexplained error unless that dependence has been explicitly modeled.

This assumption is especially important for time-series data, repeated measurements, clustered observations, and spatial data. For example, measurements collected from the same person over time may naturally be related.

Why it matters

When errors are correlated, the OLS coefficient estimates may still be useful in some settings, but standard errors and statistical tests can become incorrect. The problem is therefore particularly important when making inference.

Common warning signs

• Residuals show a sequence or time-dependent pattern.

• Repeated observations come from the same individual or group.

• Data are naturally clustered by location, organization, class, or another unit.

Assumption 3 — Homoscedasticity

Homoscedasticity means that the variance of the errors is approximately constant across the range of predicted values or predictor values. In simple terms, the spread of residuals should remain reasonably stable.

A common violation is called heteroscedasticity. In a residual-versus-fitted plot, it may appear as a funnel or cone shape, where residuals become increasingly spread out as fitted values increase.

Why it matters

Heteroscedasticity can make conventional standard errors unreliable. This can affect confidence intervals and hypothesis tests even when the estimated regression line itself appears reasonable.

How to assess it

• Plot residuals against fitted values.

• Look for changing vertical spread.

• Use formal tests when appropriate, while remembering that visual diagnostics are also important.

Possible responses

• Transform the dependent variable when scientifically appropriate.

• Use heteroscedasticity-robust standard errors for inference.

• Consider a model that explicitly represents changing variance when necessary.

Assumption 4 — Normality of Residuals

The normality assumption concerns the distribution of the regression residuals. It states that residuals should be approximately normally distributed, particularly when classical small-sample statistical inference is required.

Normality is often misunderstood. It is not an assumption that the predictor variables themselves must be normally distributed. It is the residuals that are examined.

Why it matters

With sufficiently large samples, many regression estimates are approximately normal because of large-sample behavior, so mild deviations from normality may have limited practical impact. In small samples, however, strong non-normality can affect classical inference.

How to assess it

• Inspect a histogram of residuals.

• Use a Q-Q plot to compare residual quantiles with normal-theory quantiles.

• Consider the size and nature of deviations rather than relying on a single test.

Assumption 5 — No Perfect Multicollinearity

Multicollinearity occurs when predictors contain overlapping information. The critical assumption for OLS is that there is no perfect linear relationship among the predictors.

If one predictor can be calculated exactly from other predictors, the individual coefficients cannot be uniquely estimated. This is perfect multicollinearity.

Why it matters

Strong, but not perfect, multicollinearity can still make individual coefficient estimates unstable. Standard errors may increase, coefficients may become sensitive to small changes in the data, and interpretation of individual predictors can become difficult.

How to assess it

• Inspect correlations among numeric predictors.

• Use variance inflation factor (VIF) as a diagnostic when appropriate.

• Look for redundant variables or predictors that encode nearly the same information.

4. Residuals and Model Diagnostics

A residual is the difference between an observed value and the value predicted by the regression model. Residual analysis is one of the main ways to check whether the model assumptions are reasonable.

Useful diagnostic views include residuals versus fitted values, Q-Q plots, residuals over time, and influence diagnostics. Each view answers a different question.

5. Outliers and Influential Observations

An outlier is an observation that differs substantially from the general pattern of the data. An influential observation is one that has a disproportionately large effect on the fitted regression model.

These concepts are related but not identical. A point can have a large residual without strongly changing the fitted model, while a point with unusual predictor values can strongly affect the regression coefficients.

• Check unusual observations rather than automatically deleting them.

• Investigate whether an unusual value is a data error or a genuine case.

• Use domain knowledge before deciding how to handle influential observations.

6. R^2 and Adjusted R^2

R^2, or the coefficient of determination, describes the proportion of variation in the dependent variable explained by the fitted regression model relative to a baseline model.

A higher R^2 does not automatically mean that a model is appropriate. A model can have a high R^2 while violating important assumptions, and a lower R^2 can be reasonable in noisy real-world problems.

Adjusted R^2 modifies R^2 to account for the number of predictors. It is useful when comparing models with different numbers of predictors, although it should still be interpreted alongside diagnostics and domain knowledge.

7. Bias, Variance, and Generalization

Regression models should not only fit the training data; they should also generalize to unseen data. A model that is too simple may have high bias, while an unnecessarily complex model may have high variance.

Regularization methods such as Ridge and Lasso can help control model complexity. Cross-validation is commonly used to estimate how well a model may perform on unseen data.

8. Practical Workflow for Linear Regression

• Understand the target variable and predictors.

• Clean the data and handle missing or invalid values.

• Explore relationships using visualizations and summary statistics.

• Fit an appropriate linear regression model.

• Inspect residuals and check the five assumptions.

• Investigate outliers and influential observations.

• Evaluate predictive performance on appropriate validation or test data.

• Interpret coefficients in the context of the problem.

9. Final Summary

The five assumptions provide a framework for understanding when ordinary least squares regression behaves as expected: the relationship should be appropriately linear, errors should be independent, error variance should be reasonably constant, residuals should be approximately normal when required for classical inference, and predictors should not exhibit perfect multicollinearity.

The most important practical lesson is that assumptions should be treated as diagnostic questions rather than as boxes to tick blindly. Regression is a modeling tool, and the quality of the conclusions depends on the data, the design of the study, the purpose of the model, and the evidence from diagnostics.